GASP-NE interim report on UO PM2.5 data in Newcastle City Centre

Axa-Maria Lääperi, Dr Aleksandra Svalova, Prof Philip James, Prof Anil Namdeo, Dr Graeme Sarson, Prof Anvar Shukurov

School of Mathematics, Statistics and Physics
Newcastle University

July 2026

1 Summary

The Urban Observatory (UO) has a collection of precison MONITOR sensors and MESH sensors. For use in a multi-sensor spatial model as planned for our proposed predictive online tool for the monitoring and forecast of air quality variables (pollutant concentrations) we require as many sensors as possible and therefore verification that the UO sensor data agree with that from DEFRA and AURN sensors is needed. We were made aware of NCC's restrictions in only using calibrated DEFRA-approved and -owned AURN instrumentation and locally-managed automatic monitoring sensors. The UO MESH sensors are not calibrated and not approved by DEFRA and therefore NCC cannot use then. We need to verify that the UO MONITOR sensor data agrees with that of DEFRA and AURN sensors and examine the UO MESH sensors for disagreement.

For the purposes of this report we will use the following acronyms:

- DEFRA-owned and approved AURN sensors (DEFRA)
- Locally managed automatic monitoring sensors (Local)
- UO precision MONITOR sensors (UO-Mon)
- UO MESH sensors (UO-Mesh)

We have highlighted the following as areas for priory action/attention:

- For PM2.5 data, the UO-Mon and UO-Mesh sensors have low agreement. We will proceed with using the UO-Mon sensors only.
- Some UO-Mesh and UO-Mon sensors have faulty wind direction and speed data, shown as constant over the course of a day, week and month, we have taken some decommisoned sensor data from 2025 to produce analysis.
- DEFRA Newcastle Centre sensor is systematically rounding PM2.5 values to nearest integer, this will be brought to NCC's attention.
- To validate the UO-Mon sensors we compared them to their closest DEFRA/Local sensors, all UO-Mon sensors compared had very good agreement with their closest DEFRA/Local sensor. Some UO-Mon sensors were compared with further (in distance) DEFRA/Local sensors in order to compare all DEFRA/Local sensors. We found increase in distance to change the relationship between sensors to resemble an exponenital. This was not fully consistent and we would appreciate some expert UO opinion on this matter.

The report is structured as follows:

Section Section 2 presents the locations of all DEFRA, Local and UO-Mon sensors around the Newcastle upon Tyne area that are used within this report on air quality variables (PM2.5, NOx and NO2 concentrations).

Section Section 3 presents comparisons of PM2.5 concentrations from closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest DEFRA/Local sensors to UO-Mon sensors. March 2026 reasonably demonstrates a longer time period where these sensors could differ in 15min - 1hr intervalled collection. We chose a month that was relatively recent with minimal missing sensors and values. Caveat: there is a short time period on the 21th-22st of March 2026 where data is missing from all Local sensors. The section also presents an analysis of closest UO-Mon sensors to UO-Mesh sensors over the course of a week (30th March - 5th April 2026) to discern if shorter time intervals are accurate.

Section Section 4 presents a discussion of comparisons of PM2.5 concentrations to NOx and NO2 concentrations from same sensors for DEFRA, Local and UO-Mon sensors. We chose a shorter time scale for this analysis, 30th March - 5th April 2026, as NOx and NO2 concentrations degrade very fast, so only a short time scale is appropriate for their analysis.

Section Section 5 presents a discussion on extraneous factors affecting PM2.5 concentrations over March 2025. First we examine wind effects, comparing speed and direction to concentration in 2026 was not possible as stated in the above reasoning. With 2025 data, we take old, decommisioned sensors and perform similar analysis with closest UO-Mon sensors and Local sensors. DEFRA sensors were not considered as they are situated further away. We are continuing to compare other factors such as temperature and elevation.

Table of contents

2 Sensor Locations

The three maps in Figure 1 show closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors, and closest UO-Mon sensors to UO-Mesh sensors. These pairs will be used to analyse correlations between closest sensors and assess the relibility and accuracy of the UO-Mon sensors. The distances (km) between these pairs are stated in Section 3 on the correlation analysis.

Figure 1: Maps of Newcastle upon Tyne area showing locations of (left to right) closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors, and closest UO-Mon sensors to UO-Mesh sensors.

3 Nearby Sensor Comparisons

We compare correlations in PM2.5 concentrations between the closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest DEFRA/Local sensors to UO-Mon sensors over March 2026. March was chosen as the dataset had reliable consistent data, i.e. minimal missing values and missing sensor data. However, there was still a portion of the data missing around the 21th-22nd from the Local sensors. We have also compared correlations in PM2.5 concentrations in the closest UO-Mon sensors to UO-Mesh sensors a week (30th March - 5th April 2026) to cover both long and short time interval accuracy of the UO-Mesh sensors.

3.1 Closest DEFRA/Local to UO-Mon sensors

Figure 2 presents the data in the form of scatter plots and time series, where only the closest UO-Mon sensor is included for each DEFRA/Local sensor. The aim of this exercise is to assess the quality and accuracy of the UO-Mon sensor data. Both representations of the data demonstrate a very good agreement between the sensors. Figure 3 shows a similar comparison, but for more widely separated sensors. As might be expected we observe a worse agreement. A summary table of correlational information and distances between sensors is given in Table 1 from which we see high correlation between all closest sensor pairs. We use both Pearson's and Spearman's correlation coefficents. Pearson's evaluates linear relationships and proportional change between variables while Spearman's monotonic relationships which dicerns if variables are changing together, not necessarily at a constant rate as Pearson's would let us assume. Figure 4 also demonstrates the relationship between distance and the correlation coefficents, and lower correlations are observed when the sensor separation is larger, as expected.

Other relationship (nonlinear) might also exist; we have examined other external factors (such as temperature, wind effects, etc,) in Section 5.

In Figure 3, we also examined the agreement between Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge sensors since we have a comparison with UO-Mon Gateshead Tyne Bridge (in the same location). This way we can deduce if the disagreement is indeed related to distance rather than any problems with the sensor quality and accuracy. We see a similar nonlinear relationship occuring between these two sensors as in the case of the Local-Gateshead A1 Dunston and UO-Mon Gateshead Tyne Bridge sensors, which indicates this nonlinear relationship is mainly due to the distance between the sensors.

As shown in Table 1 and Figure 4, the data from the DEFRA and UO-Mon sensors are very consistent, with the correlation coefficients exceeding 0.7 even when the separation of the sensors is as large as 2-3 km. This also suggest the presence of a background PM2.5 distribution which varies little over the distance of a few kilometers. This background and its spatial and temporal variations can thus be identified using the data available.

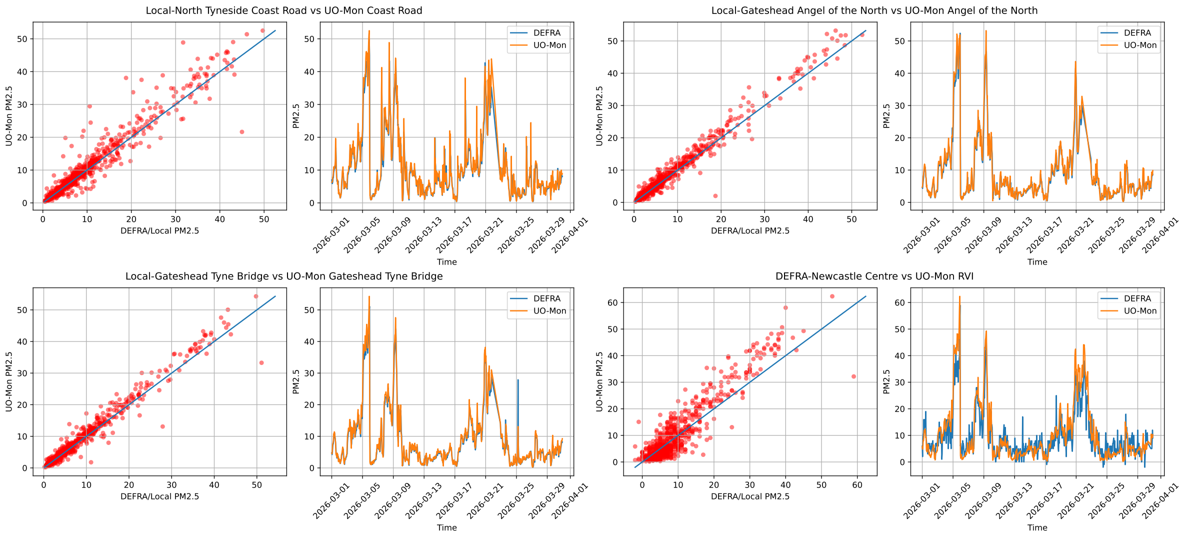

Figure 2: Comparison of PM2.5 concentrations measured by each DEFRA/Local sensor and its closest UO-Mon sensor in March 2026. Only the closest UO-Mon sensor is included for each DEFRA/Local sensor. We see very good agreement between the sensors.

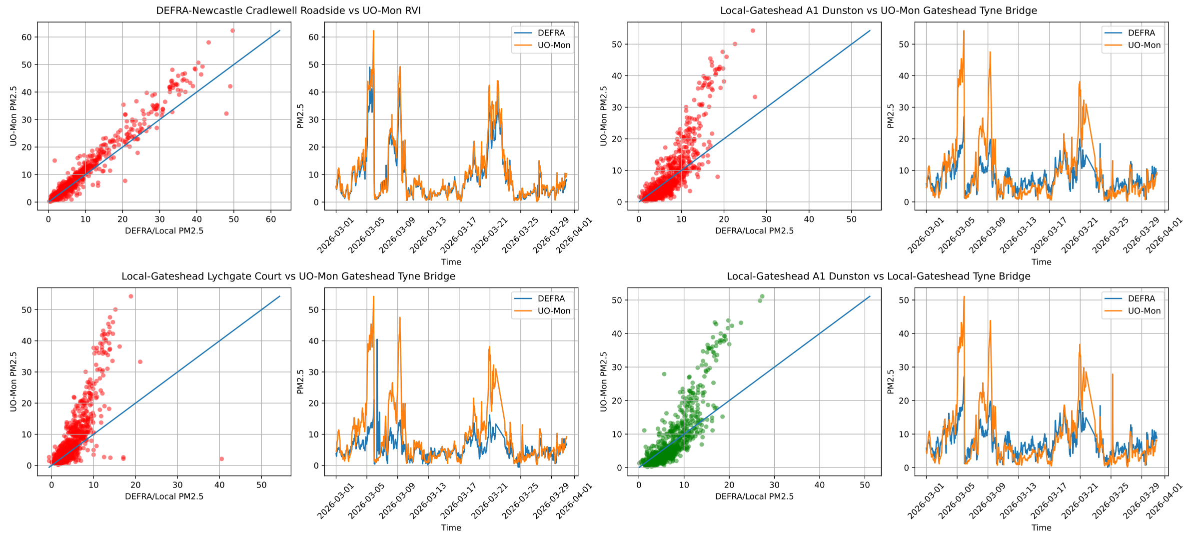

Figure 3: Comparison of PM2.5 concentrations measured by DEFRA/Local sensors and additional nearby UO-Mon sensors in March 2026. These comparisons exclude the nearest UO-Mon sensor shown in Figure 2 and illustrate relationships with other, more distant UO-Mon sensors. We see worse agreement with some nonlinear relationships which is expected due to the distance. The lower-right plot (green scatter) shows PM2.5 concentrations measured by Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge sensors. We expect to see a similar nonlinear relationship to that comparing Local-Gateshead A1 Dunston and UO-Mon Gateshead Tyne Bridge.

Table 1: Distances between sensors in km, Pearson's and Spearman's correlation coefficent for UO-Mon and closest DEFRA/Local sensors PM2.5 concentrations over long time interval. High correlation coefficents for all close sensors.

DEFRA/Local	UO-Mon	Distance km	Pearsons coeff	Spearmans coeff
Local-Gateshead A1 Dunston	UO-Mon Gateshead Tyne Bridge	3.37	0.846	0.834
DEFRA-Newcastle Cradlewell Roadside	UO-Mon RVI	1.643	0.974	0.95
Local-Gateshead Lychgate Court	UO-Mon Gateshead Tyne Bridge	0.572	0.741	0.789
DEFRA-Newcastle Centre	UO-Mon RVI	0.468	0.926	0.812
Local-Gateshead Tyne Bridge	UO-Mon Gateshead Tyne Bridge	0.006	0.983	0.971
Local-Gateshead Angel of the North	UO-Mon Angel of the North	0.003	0.986	0.962

Table 1: Distances between sensors in km, Pearson's and Spearman's correlation coefficent for UO-Mon and closest DEFRA/Local sensors PM2.5 concentrations over long time interval. High correlation coefficents for all close sensors.

DEFRA/Local	UO-Mon	Distance km	Pearsons coeff	Spearmans coeff
Local-North Tyneside Coast Road	UO-Mon Coast Road	0.002	0.961	0.954

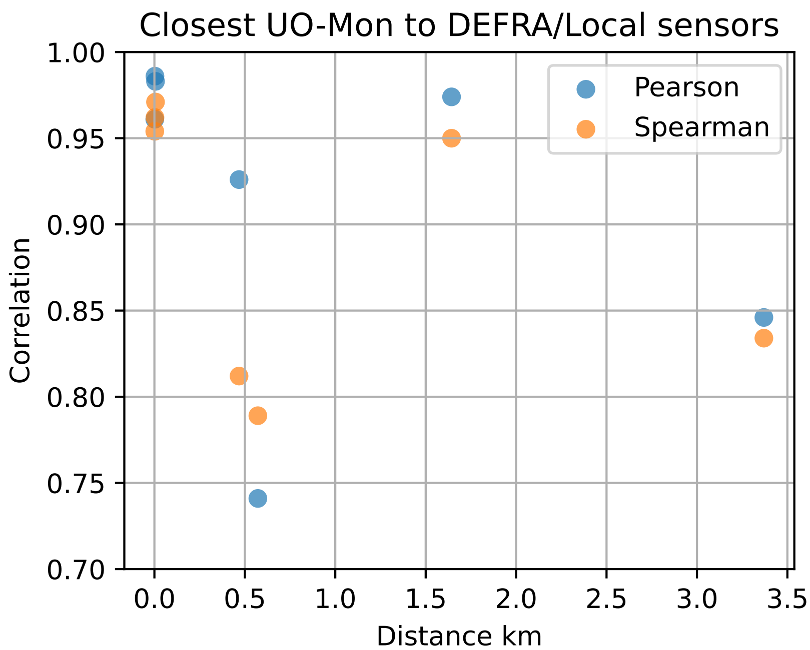

Figure 4: Correlation coefficents of each sensor pair from Figure 2 and Table 1 with their distance in kilometers.

3.2 Closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors

Using data from March 2026, we compare closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors PM2.5 concentrations. Figure 5 presents the data in the form of scatter plots for each pair of nearby sensors. The aim of this exercise is to assess the quality and accuracy of the UO-Mesh sensor data against any precision sensors. The representation of the data demonstrate a very poor agreement between the sensors which implies little to no reliability of the UO mesh sensors.

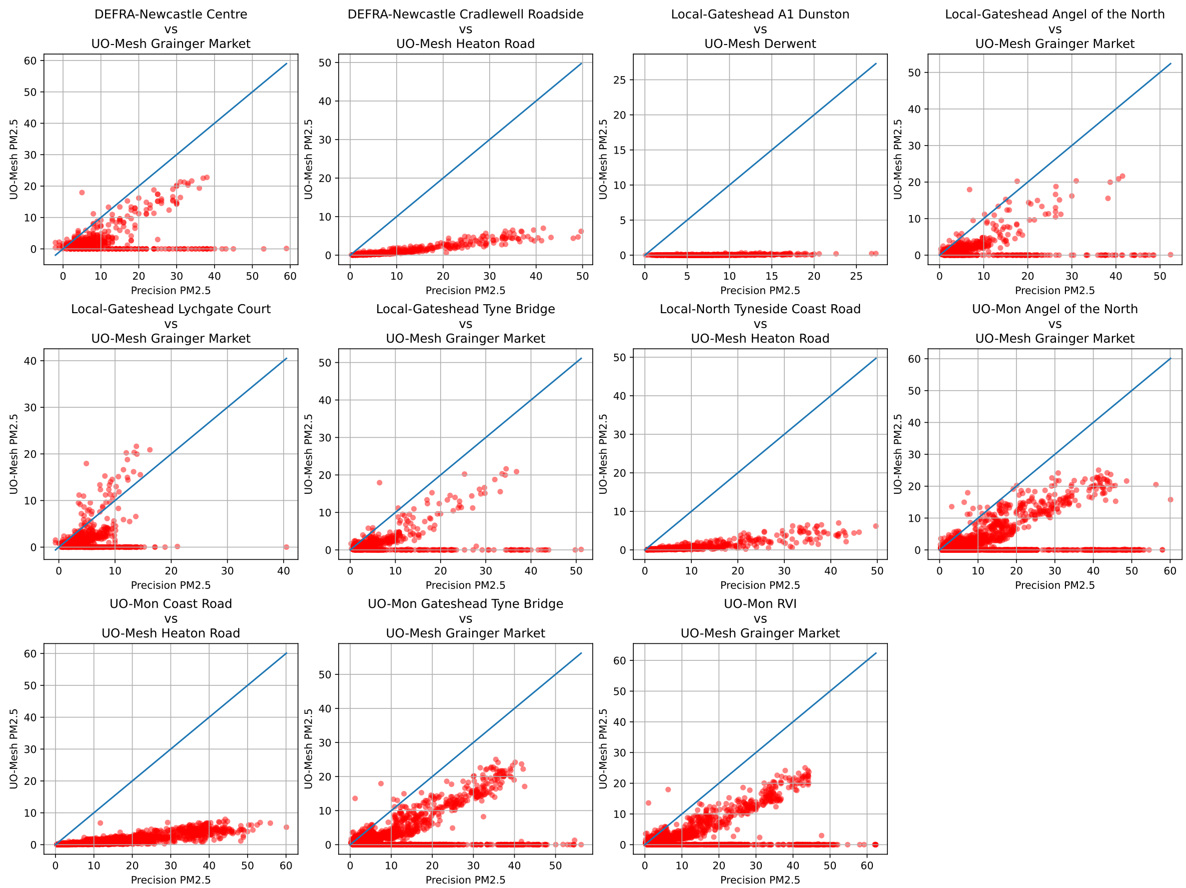

Figure 5: Comparisons of PM2.5 concentrations measured by UO-Mesh and closest DEFRA/Local/UO-Mon sensors in March 2026. We see poor agreement between the sensors.

3.3 Closest UO-Mon to UO-Mesh sensors

Using data from 30 March to 5 April 2026, we compare PM2.5 concentrations recorded by nearby UO-Mon and UO-Mesh sensors. Figure Figure 6 presents scatter plots for each sensor pair. As in Section 3.2, the aim is to assess the agreement between the UO-Mesh and UO-Mon measurements, focusing here on a shorter time interval.

Some UO-Mesh sensors show reasonable agreement with nearby UO-Mon sensors, particularly UO-Mesh West Denton and UO-Mesh Coast Road when compared with UO-Mon Coast Road. Although most other pairs show limited agreement in the measured concentrations, several exhibit moderate positive correlations. UO-Mesh Heaton Road and UO-Mesh Grainger Market have particularly high Pearson and Spearman correlations with both UO-Mon sensors, as shown in Table 2. Interestingly, UO-Mesh West Denton agrees more closely with UO-Mon Coast Road than with the geographically closer UO-Mon RVI sensor; UO-Mesh Grainger Market shows a similar pattern. The comparison between UO-Mesh Wingrove and UO-Mon RVI has a very low Pearson correlation but a moderate Spearman correlation, suggesting a possible monotonic, but not necessarily linear, relationship between their PM2.5 measurements. Overall, the data contain some promising common structure, although the strength of agreement varies considerably between weeks. The persistence of broadly similar correlation patterns across sensor pairs warrants further investigation.

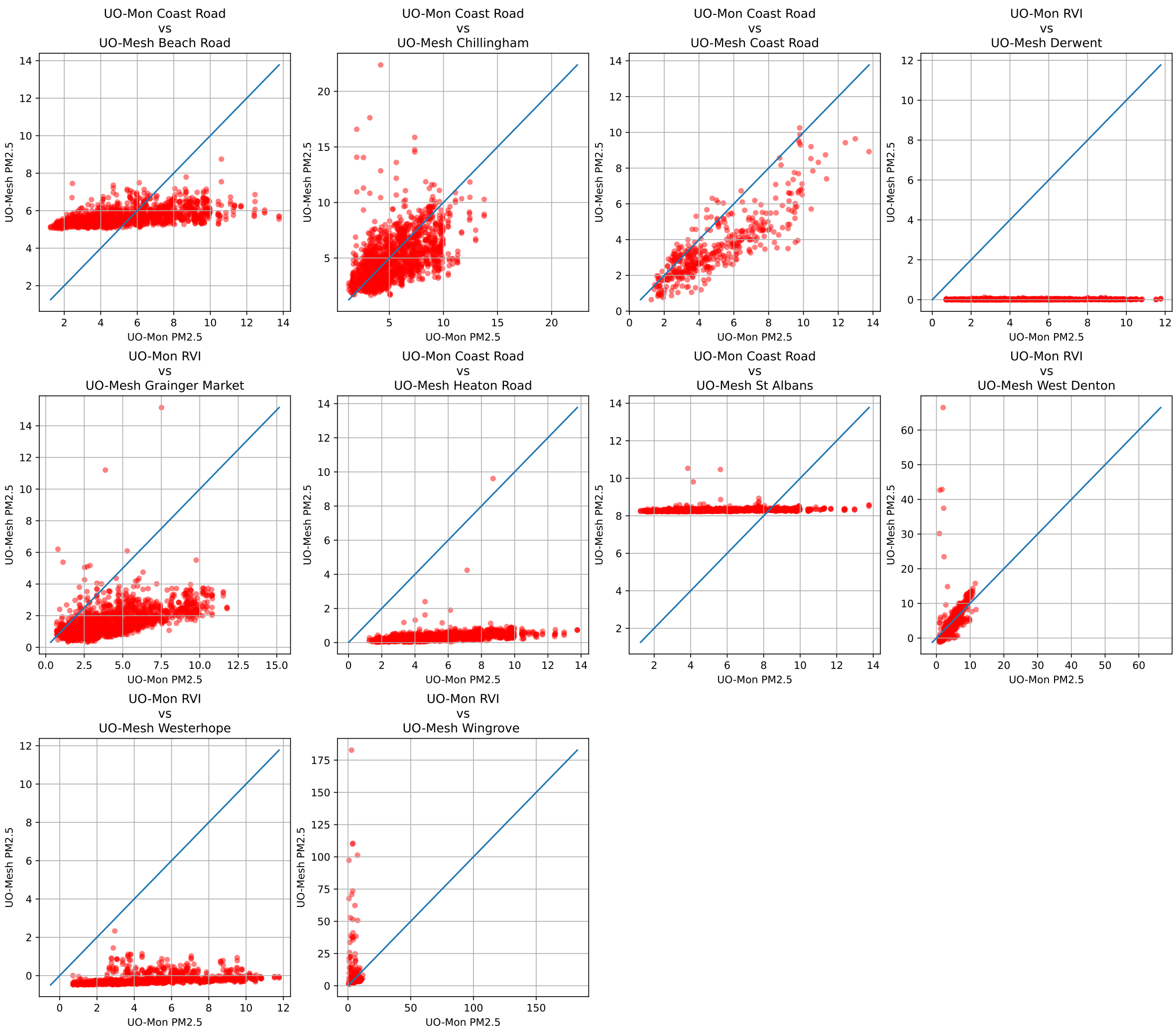

Figure 6: Comparisons of PM2.5 concentrations comparisons between UO-Mesh and closest UO-Mon sensors over short time interval. We see varied agreement between the sensors.

Table 2: Distances between sensors in km, Pearson's and Spearman's correlations for UO-Mesh and closest UO-Mon sensors PM2.5 concentrations over short time interval. We see varied correlation levels between close sensors.

UO-Mesh	UO-Mon	Distance km	Pearsons corr	Spearmans corr
UO-Mesh Coast Road	UO-Mon Coast Road	0.033	0.848	0.836
UO-Mesh Chillingham	UO-Mon Coast Road	1.757	0.544	0.615
UO-Mesh Grainger Market	UO-Mon RVI	0.754	0.538	0.64
UO-Mesh Beach Road	UO-Mon Coast Road	6.15	0.531	0.59
UO-Mesh West Denton	UO-Mon RVI	6.098	0.479	0.807
UO-Mesh Heaton Road	UO-Mon Coast Road	1.709	0.401	0.672
UO-Mesh Westerhope	UO-Mon RVI	6.045	0.385	0.691
UO-Mesh St Albans	UO-Mon Coast Road	1.904	0.231	0.499
UO-Mesh Derwent	UO-Mon RVI	5.252	0.151	0.165
UO-Mesh Wingrove	UO-Mon RVI	1.985	-0.014	0.519

4 Nitric data

We compare the PM2.5 and NOx/NO2 concentrations in the same sensors for all DEFRA, Local and UO-Mon sensors using the data from March 30 to April 5, 2026. We chose nitric emissions as a proxy of traffic influence to check whether this has an effect on PM2.5. A shorter time frame was chosen because nitric contamination decays with time occuring much faster than that of PM2.5.

4.1 DEFRA sensors

Figure 7 presents the scatter plots of the PM2.5 and NOx and No2 concentrations measured with the DEFRA sensors, and the legends show the Pearson correlation coefficient. It is clear that the two variable are not well correlated and thus are likely to depend on different factors.

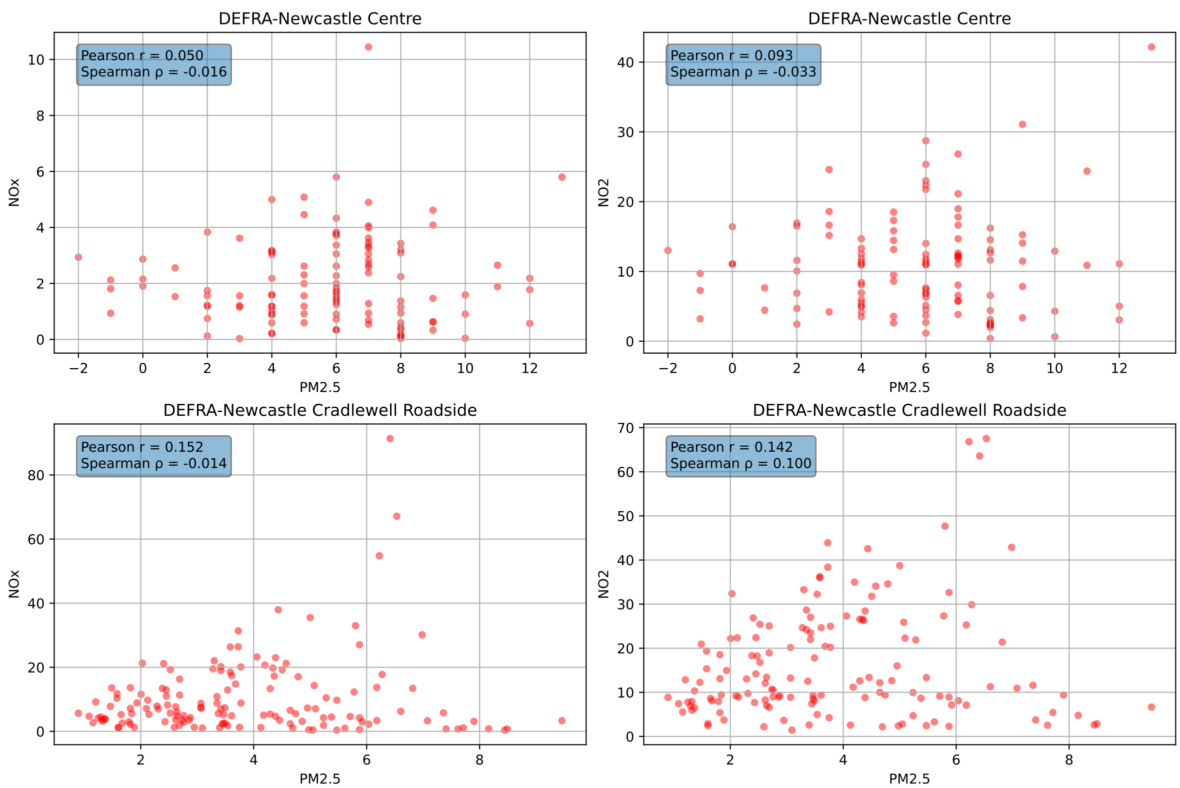

Figure 7: Comparsion of PM2.5 and NOx and NO2 concentrations for DEFRA sensors over short time interval (a week).

4.2 Local sensors

Figure 8 presents in a similar manner NO2 and PM2.5 data for the Local sensors (which do not collect NOx data). The correlation coefficient is rather low and moderate at most. This suggest that, at the location of those sensors, there may be some traffic affecting PM2.5 concentration. It cannot be excluded, however, that both NO2 and PM2.5 depend on other factors, such as the air temperature rather than being causally related.

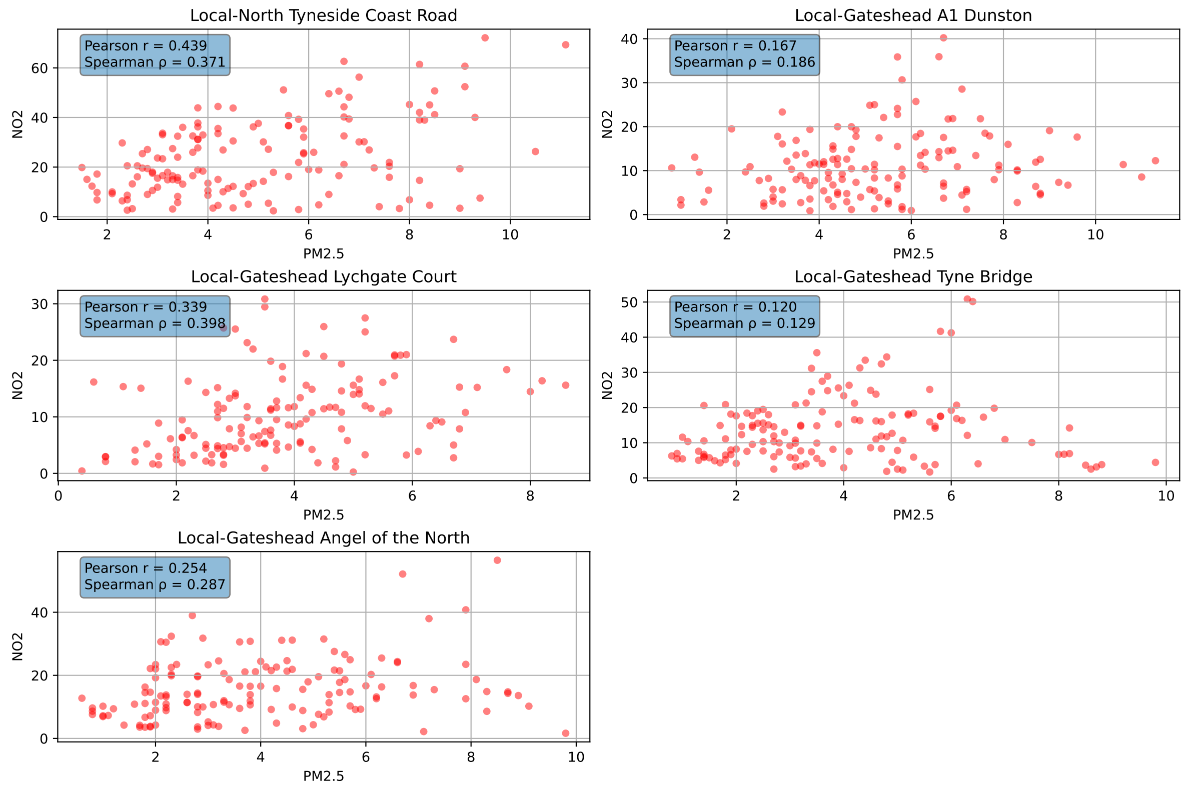

Figure 8: Comparisons of Local sensor PM2.5 and NO2 concentrations over a short time inteval (a week).

4.3 UO-Mon sensors

Figure 9 presents similar data for the UO-Mon sensors. As for the other sensors, the correlation is moderate at most and varies strongly between the sensors.

We conclude that the nitric and particulate air contaminations are poorly correlated and one cannot be used as a proxy for the other. In particular, any machine learning model used to forecast one of those variables would not benefit from the input of the measurements of the other one in the training set.

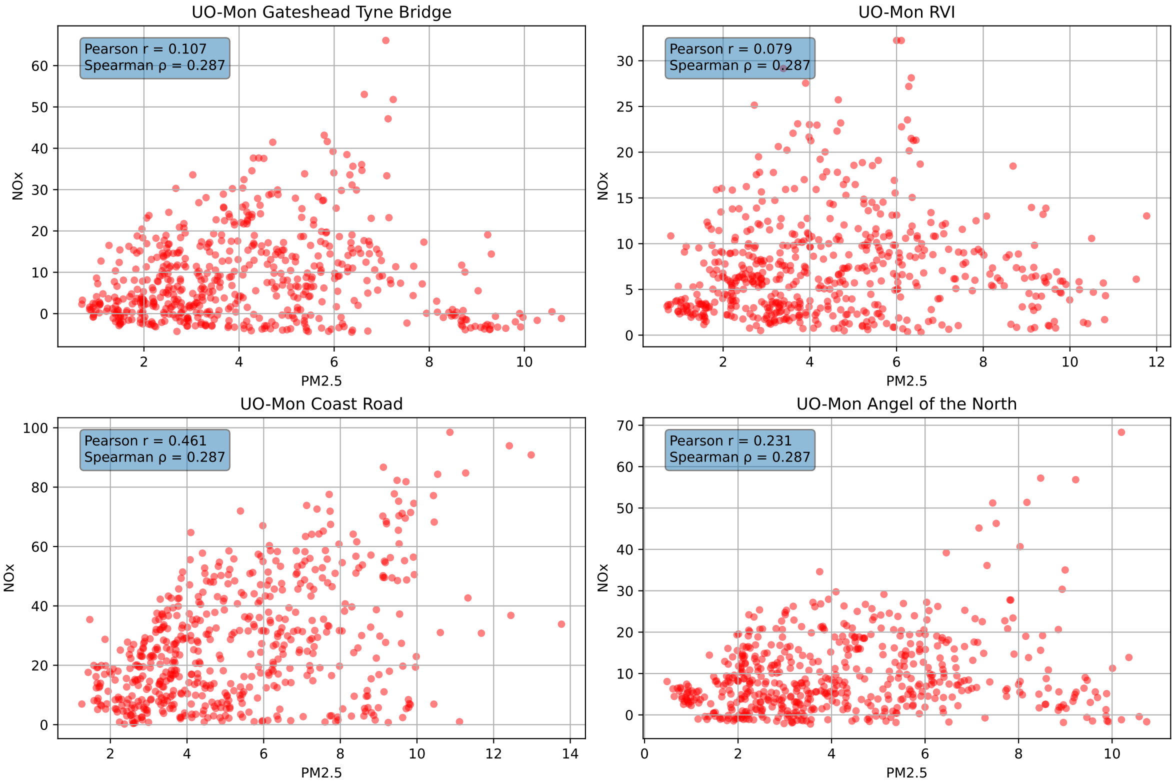

Figure 9: Comparisons of UO-Mon sensor PM2.5 and NOx concentrations over a short time interval (a week).

5 Attributes affecting PM2.5 concentraions

Environmental and geographical factors can influence both the formation and spatial distribution of PM2.5. This section examines the relationships between PM2.5 concentrations and three potentially important attributes: wind conditions, altitude and temperature.

5.1 Wind direction and speed

Wind data from the UO-Mon sensors were obtained for March 2026. However, the data appears to be faulty: as shown in Figure 10, the recorded wind speed and direction remain constant throughout the day, week and month. We therefore also examined wind measurements from two older UO sensors located in Birtley and West Denton to investigate whether any relationships with PM2.5 concentrations could be identified.

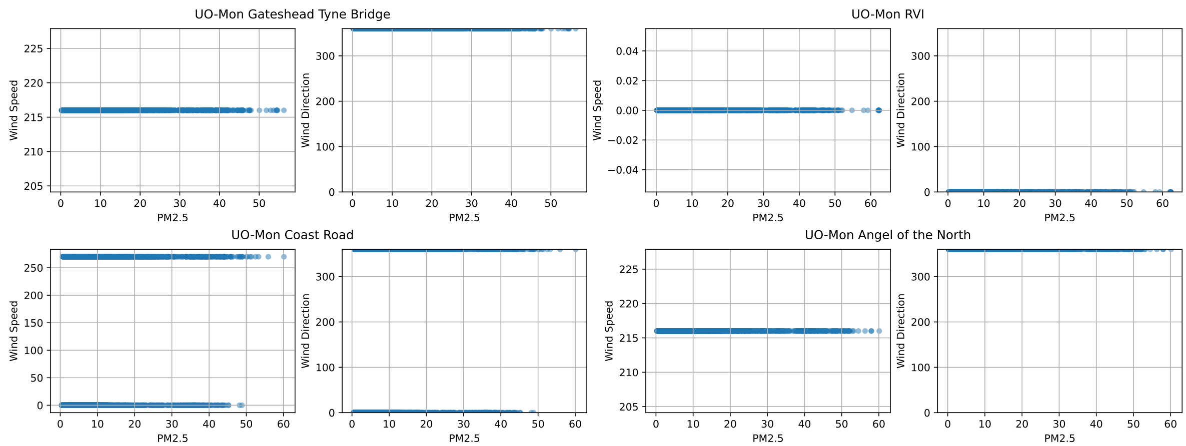

Figure 10: Wind speed and direction comparisons to PM2.5 concentrations from the UO-Mon sensors.

The two UO wind sensors located in Birtley and West Denton appear to have recorded data throughout 2025. Consistent with the previous analyses, we focus on March 2025 and examine how PM2.5 concentrations recorded by the nearest UO-Mon and Local sensors vary with wind speed and direction. Figure Figure 11 shows the locations of the wind sensors and their nearest corresponding PM2.5 sensors.

Figure Figure 12 presents three comparisons for the nearest UO-Mon sensors: scatter plots of PM2.5 concentration against wind speed, coloured by wind direction; pollution rose plots showing PM2.5 concentration by wind direction; and binned wind-speed and wind-direction plots, coloured by PM2.5 concentration. Figure Figure 13 presents the equivalent comparisons for the nearest Local sensors.

The pollution rose plots indicate that higher PM2.5 concentrations occur most frequently when the wind direction is between 0° and 90°, with some overlap into the neighbouring directional sectors. This pattern is less pronounced for the Local sensors, although a similar trend remains visible. The difference may partly reflect the lower temporal resolution of the Local PM2.5 measurements, which are recorded hourly, compared with the 15-minute measurements from the UO-Mon sensors.

The scatter and binned plots also indicate a relationship between PM2.5 concentration and wind speed. In general, lower PM2.5 concentrations are observed more consistently at higher wind speeds, which is consistent with increased dispersion of particulate matter under windier conditions.

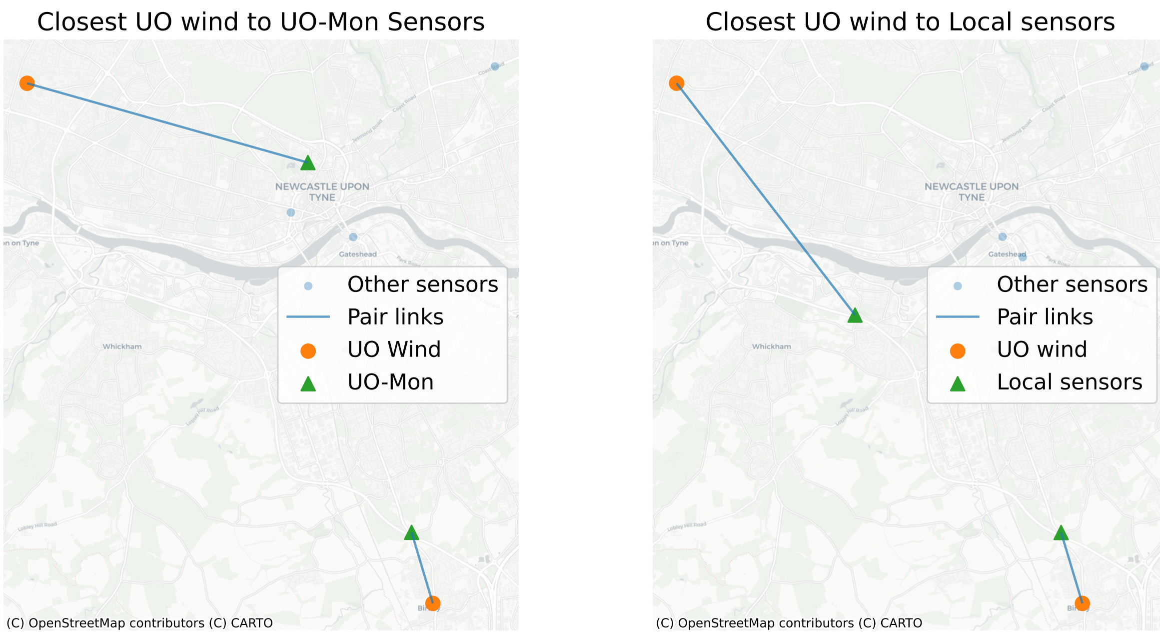

Figure 11: Maps of UO-Mesh sensors and Local sensors to UO wind sensors in Birtley and West Denton.

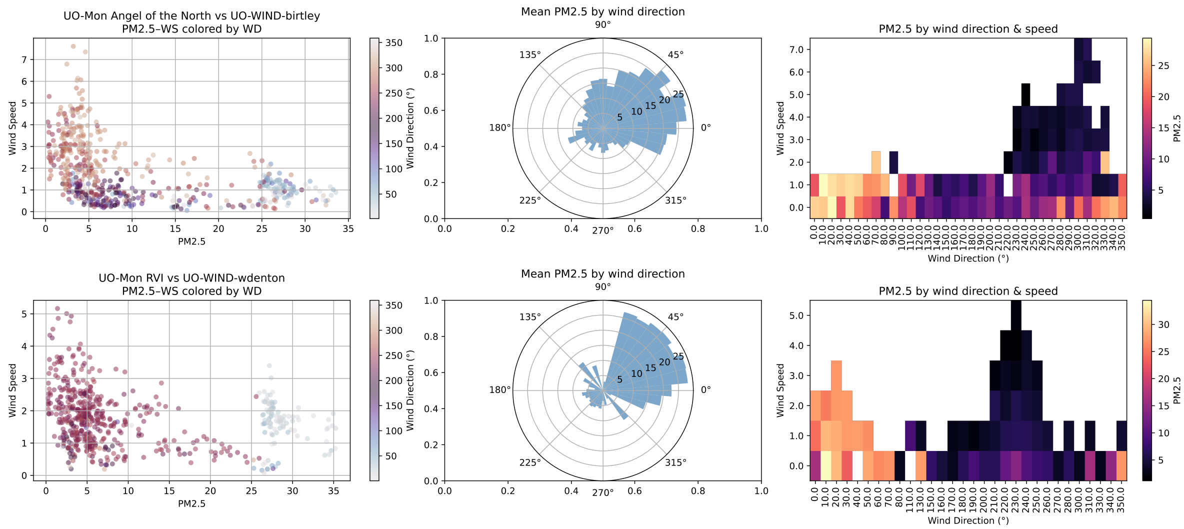

Figure 12: Scatter plots of PM2.5 concentration to wind speed coloured by wind direction, rose plots of PM2.5 concentration to wind direction, and binned plots of wind speed and direction coloured by PM2.5 concentration for the closest UO-Mon sensors to UO wind sensors in Birtley and West Denton.

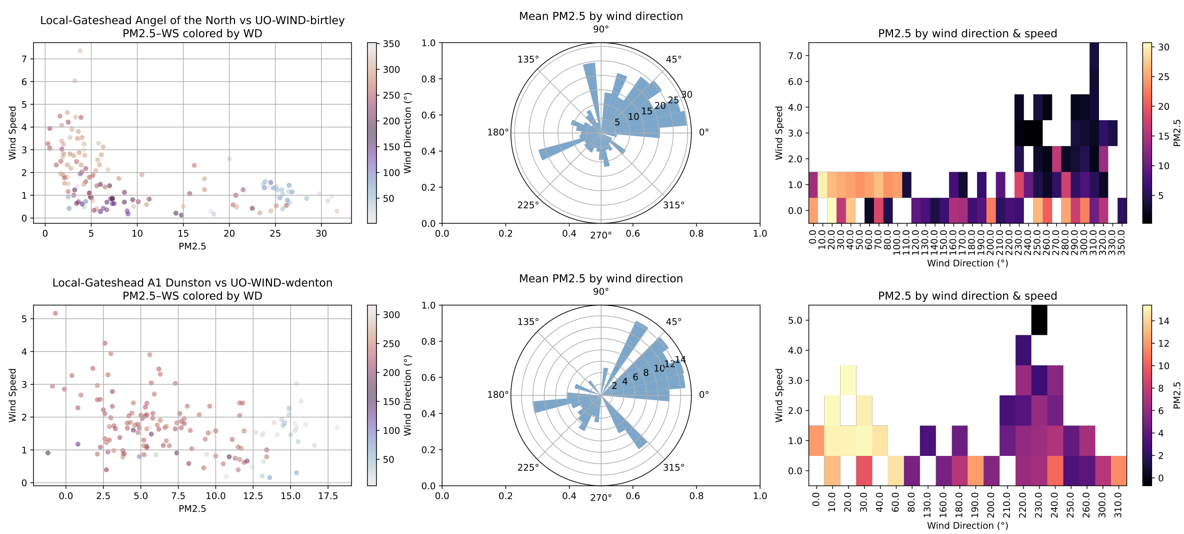

Figure 13: Scatter plots of PM2.5 concentration to wind speed coloured by wind direction, rose plots of PM2.5 concentration to wind direction, and binned plots of wind speed and direction coloured by PM2.5 concentration for the closest Local sensors to UO wind sensors in Birtley and West Denton.

5.2 Altitude

An Ordnance Survey digital elevation model covering the Newcastle upon Tyne DEFRA, Local and UO-Mon sensor network was used to investigate whether PM2.5 concentrations vary with sensor altitude.

Figure Figure 14 compares the altitude of each sensor with the mean and standard deviation of its PM2.5 measurements over the month. A weak positive correlation is observed between altitude and mean PM2.5 concentration, while a moderate positive correlation is observed between altitude and the variability of the measurements. This suggests that sensors at higher altitudes may experience greater variability in PM2.5 concentrations, although further analysis would be required to determine whether altitude itself is responsible for this relationship.

To investigate whether any temporal patterns were associated with altitude, we also examined one week of PM2.5 time-series data, with sensors distinguished by altitude in Figure 15. However, there are few clear differences in the temporal patterns between sensors at different elevations.

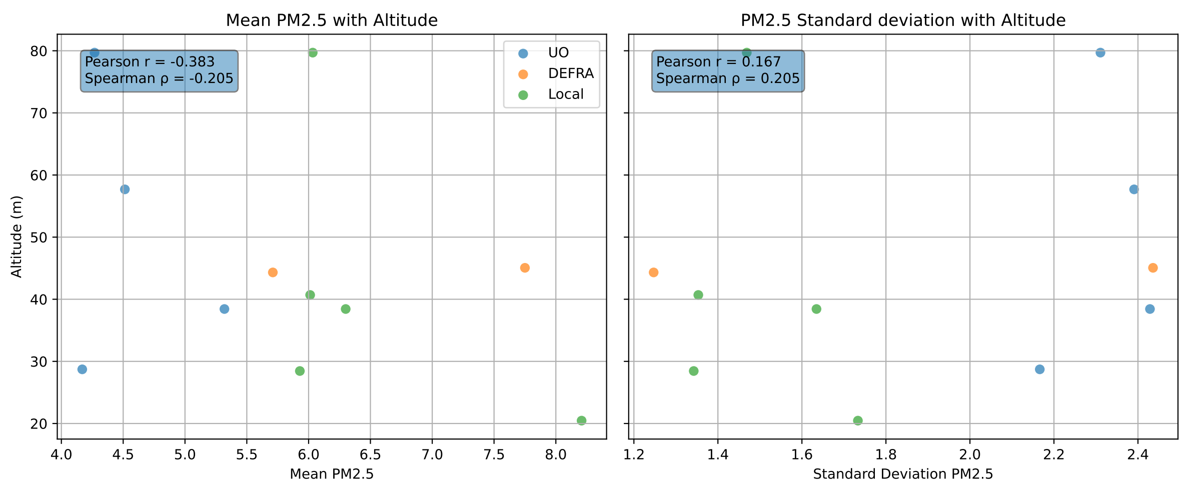

Figure 14: Altitude of DEFRA/Local/UO-Mon sensors with mean and standard deviation in PM2.5 concentration over long time interval.

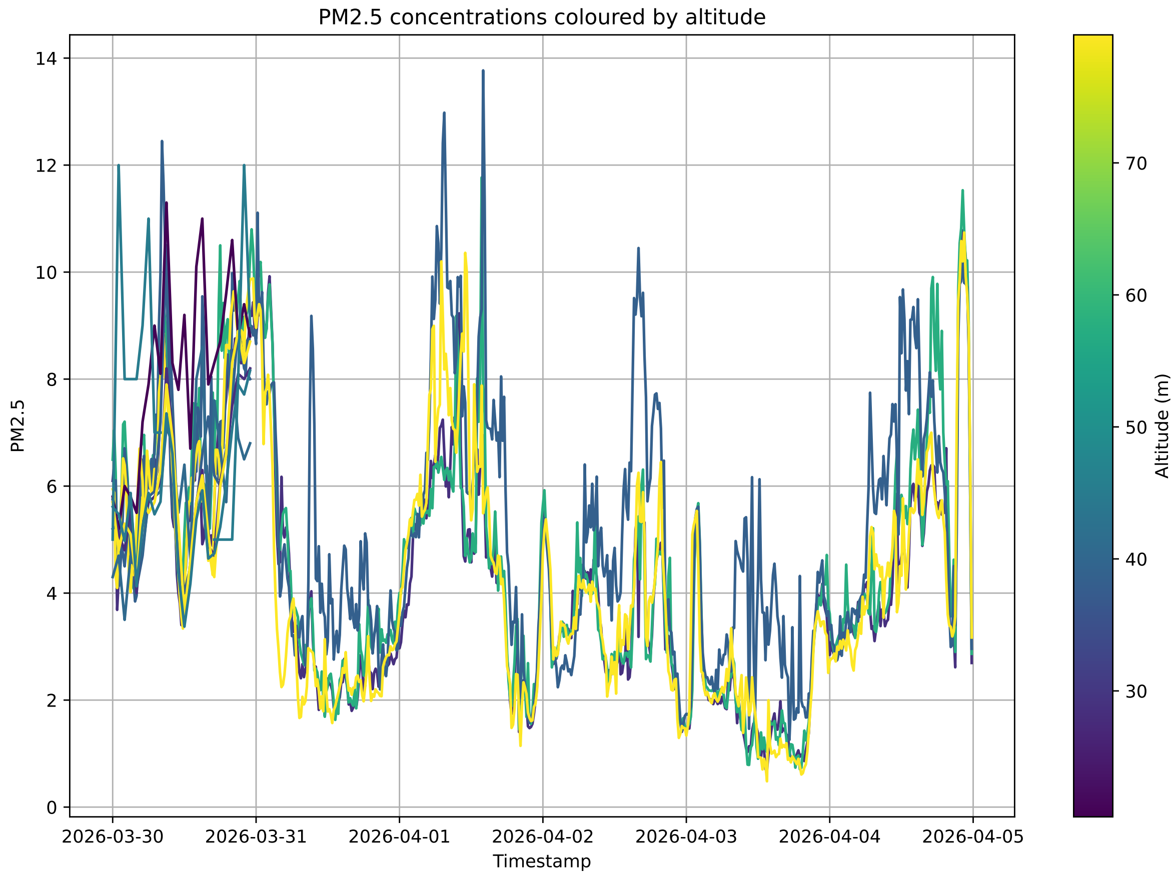

Figure 15: Timeseries of PM2.5 concentrations for different sensors coloured by altitude.

5.3 Temperature

PM2.5 concentrations recorded by the UO-Mon sensors were compared with temperature measurements from the same sensors over March 2026. This analysis was undertaken to investigate whether temperature may help explain variations in PM2.5 concentrations, alongside the relationships with NOx and NO2 considered previously.

Figure 16 presents scatter plots of PM2.5 concentration against temperature for each UO-Mon sensor. The absence of a clear pattern in these plots, together with the low correlation coefficient, indicates little evidence of a relationship between temperature and PM2.5 concentrations over the monthly study period.

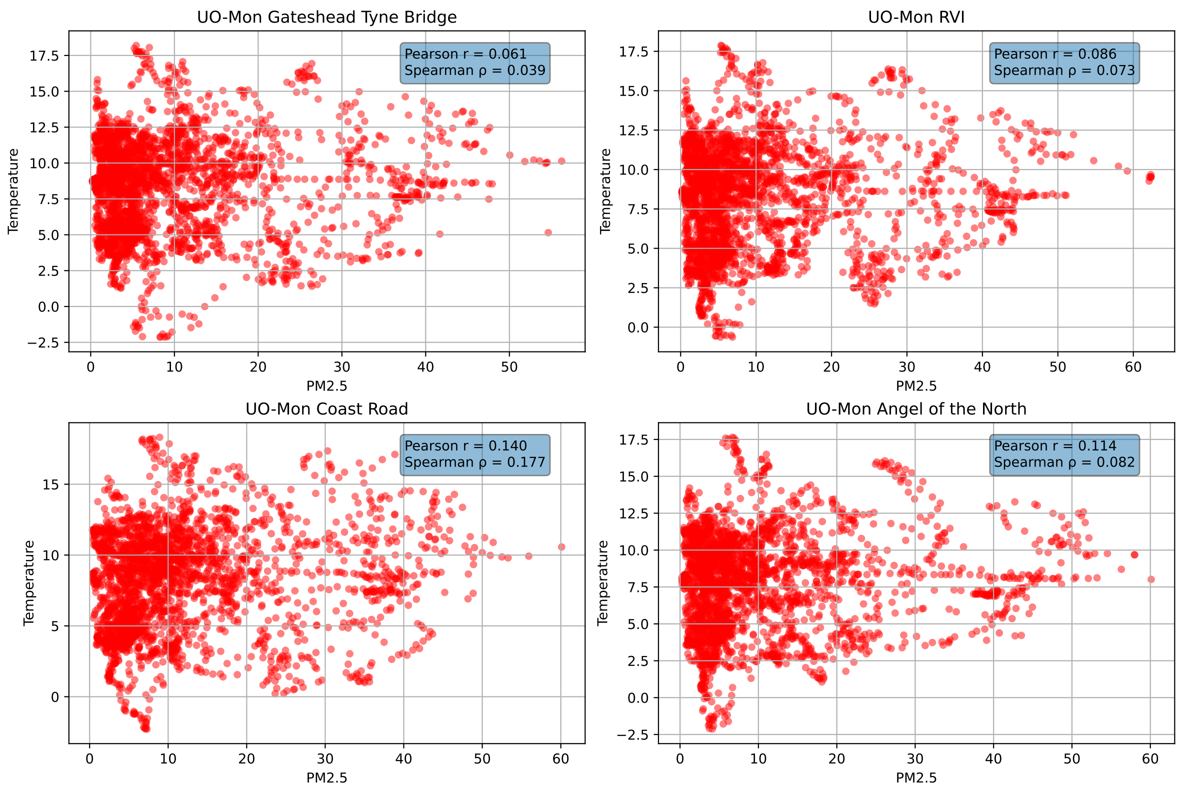

Figure 16: Comaprsion on PM2.5 concentration and temperature from UO-Mon sensors.

6 Conclusions

The analyses presented in this report demonstrates that the DEFRA, Local and UO-Mon sensors exhibit sufficient agreement to support their combined use in further modelling. We investigated the influence of external factors, including wind conditions, temperature and sensor altitude, to provide further context for the PM2.5 measurements and inform a predictive machine learning model. Based on this, we suggest to proceed using the DEFRA, Local and UO-Mon sensors to develop a multi-sensor spatial Gaussian Process (GP) model, which will form the basis of the proposed predictive online tool. The inclusion of the UO-Mon sensors would significantly extend the data base for the air quality forecast and allow us to achieve higher reliability and accuracy of the predictions.